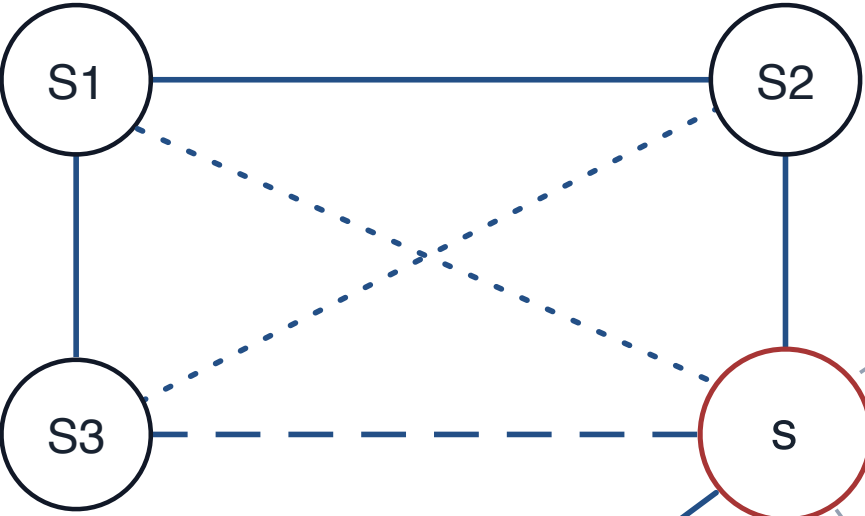
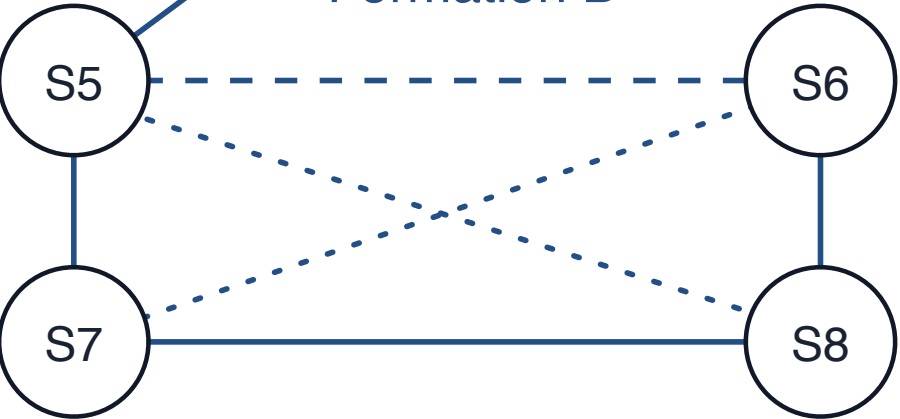


Distributed formation-tracking setting

Formation A



Formation B



— high-reliability
- - moderate
... lossy

Representative node s local processing

s expanded local view of node s

Evidence available at node s

Local measurements
measurement update at node s

Local LMB posterior
kinematic states and existence

Neighbor information
neighbor posteriors
and link outcomes

Branch-decoupled adaptive fusion

Quality cues

posterior precision, link reliability,
existence confidence

Shared quality score
common evidence

Spatial path

Gaussian KLA

Existence path

Bernoulli KLA

FID-FIA cue

Fused local posterior

extraction and forwarding